

↳ Polynomial Representation.

↳ any polynomial of degree d

↳ represented in $d+1$ points.

↳

$$P(x) = p_0 + p_1 x + p_2 x^2 + p_3 x^3 + \dots + p_d x^d.$$

$$P(x_0) = p_0 + p_1 x_0 + p_2 x_0^2 + p_3 x_0^3 + \dots + p_d x_0^d.$$

1

$\vdots$

$$P(x_d) = p_0 + p_1 x_d + p_2 x_d^2 + p_3 x_d^3 + \dots + p_d x_d^d.$$

$$\begin{bmatrix} P(x_0) \\ P(x_1) \\ \vdots \\ P(x_d) \end{bmatrix} = \begin{bmatrix} 1 & x_0^1 & x_0^2 & x_0^3 & x_0^d \\ 1 & x_1^1 & x_1^2 & x_1^3 & x_1^d \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_d^1 & x_d^2 & x_d^3 & x_d^d \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_d \end{bmatrix}$$

$\underbrace{\hspace{15em}}_M$

↳ M is always invertible for unique $x_0, x_1, \dots, x_d$.

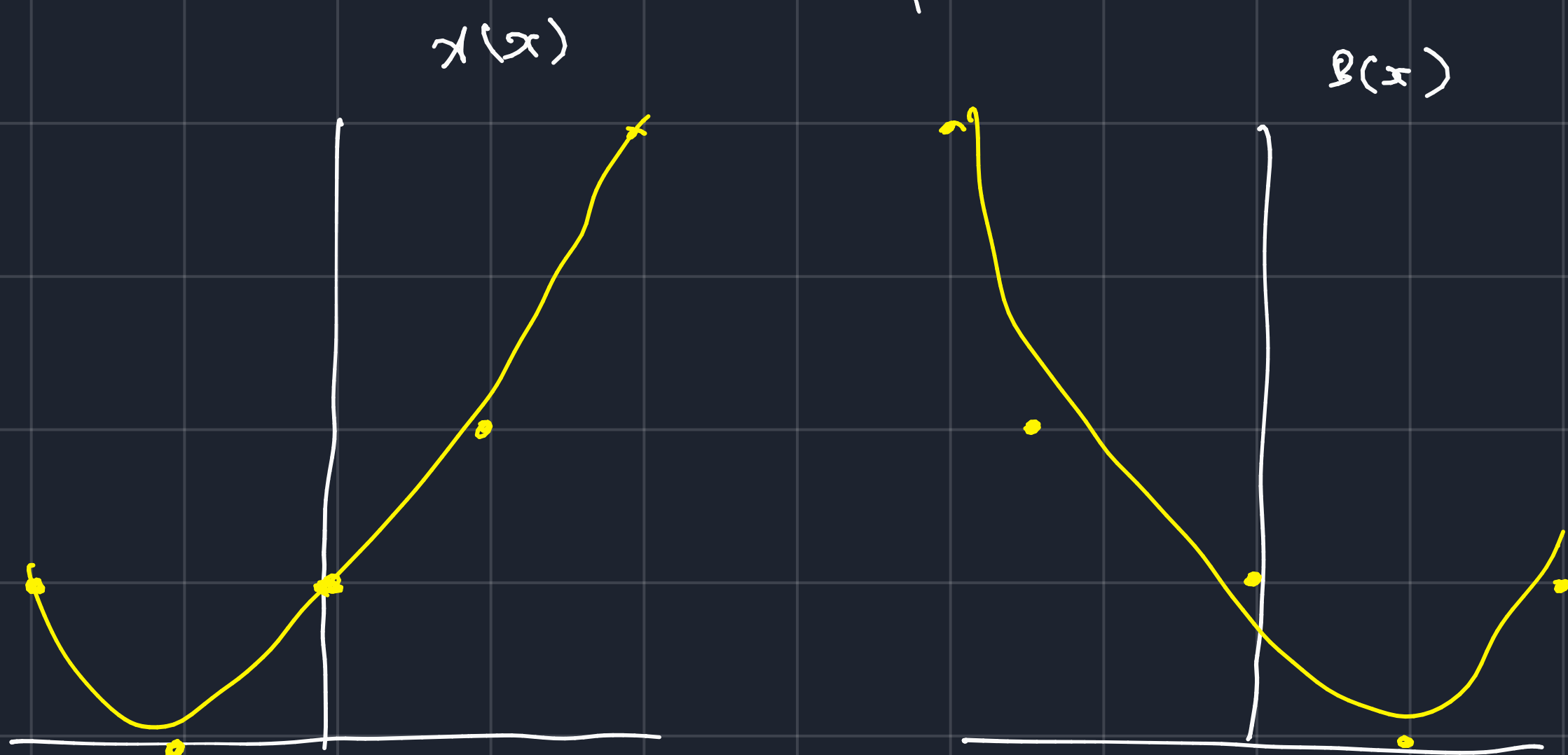
$$A(x) = x^2 + 2x + 1$$

$$B(x) = x^2 - 2x + 1$$

$$C(x) = A(x) \cdot B(x)$$

↳ Degree 4 → points to → 5

represent



$$[(-2, 1), (-1, 0), (0, 1), (1, 4), (2, 9)] \times [(-2, 9), (-1, 4), (0, 1), (1, 0), (2, 1)]$$

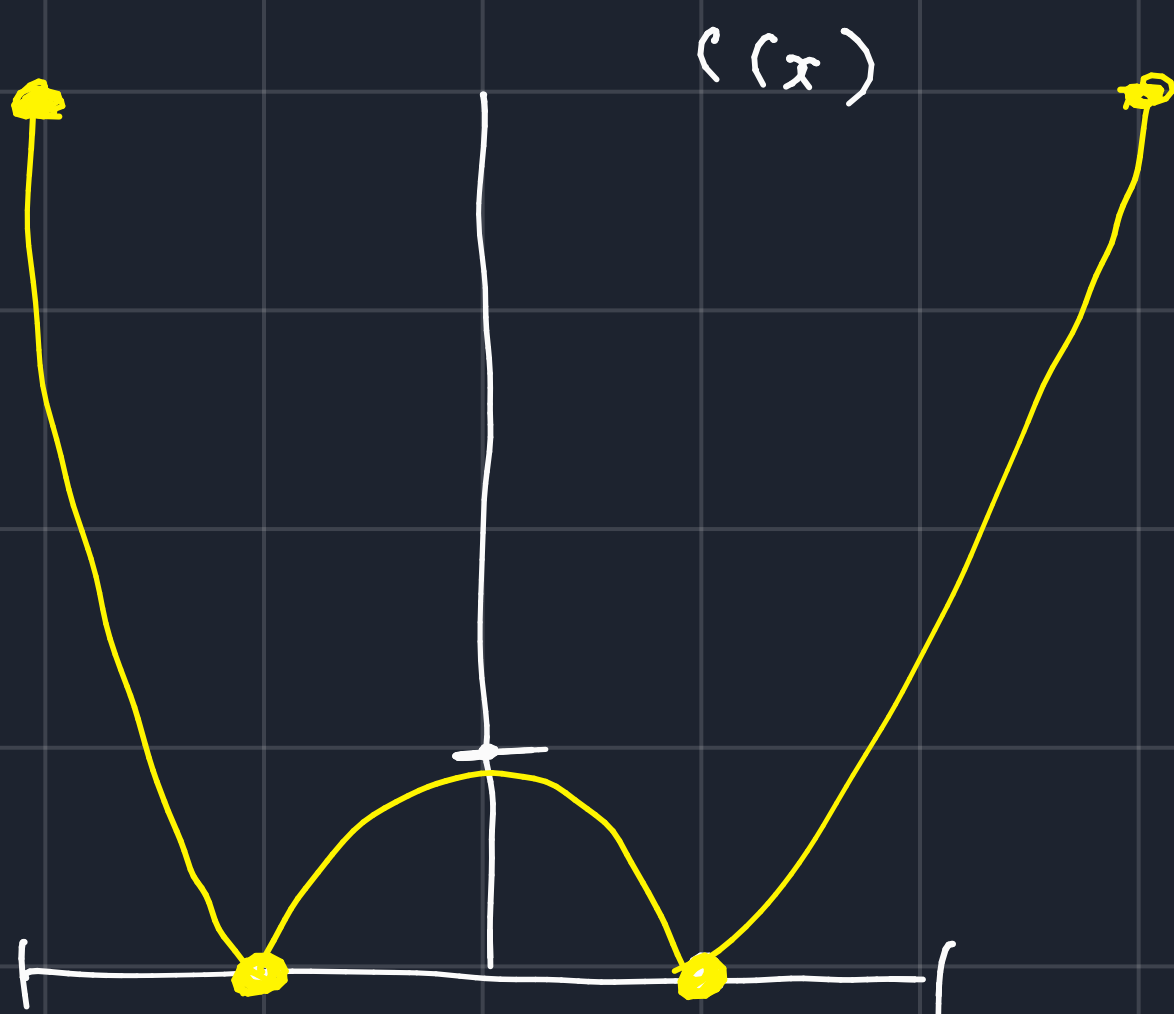
= value of C plots.

→ how to multiply?

↳ keeping the common x-values

↳ multiply the y-part

$$[(-2, 9), (-1, 0), (0, 1), (1, 0), (2, 9)]$$



$$C(x) = x^4 - 2x^2 + 1$$